

Ringkasan Dataset

Dataset ini berisi 254.736 baris data kualitas udara per jam dari 29 kota besar di dunia (termasuk Lahore, Delhi, Beijing, Jakarta, dll.) selama periode Mei 2025 – Mei 2026.

Kolom yang tersedia:

Timestamp — waktu per jam

City, Latitude, Longitude — identitas lokasi

PM2_5_ug_m3, PM10_ug_m3 — partikel polutan utama

Carbon_Monoxide_ug_m3, Nitrogen_Dioxide_ug_m3, Ozone_ug_m3, Dust_ug_m3 — gas polutan

UV_Index — indeks UV

European_AQI — indeks kualitas udara

Hazardous_Event — apakah terjadi event berbahaya (0/1)

Tujuan

yaitu Memprediksi Regresi — "Berapa nilai PM2.5 jam berikutnya?" Memprediksi angka konsentrasi polutan di masa depan.

PM2.5 adalah partikel udara yang sangat kecil berukuran 2.5 mikrometer atau lebih kecil.

1.Import And Read Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.patches as mpatches
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import GridSearchCV, TimeSeriesSplit
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
df = pd.read_csv('global_urban_smog_pm25_hourly.csv')
df.head()
```

	Timestamp	City	Latitude	Longitude	PM10_ug_m3	PM2_5_ug_m3	Carbon_Monoxide_ug_m3	Nitrogen_Dioxide_ug_m3	Ozone_ug_m3	D
0	2025-05-22T00:00	Lahore	31.5204	74.3587	186.7	74.3	926	22.9	94	
1	2025-05-22T01:00	Lahore	31.5204	74.3587	286.8	96.4	736	18.1	103	
2	2025-05-22T02:00	Lahore	31.5204	74.3587	380.0	122.0	599	14.1	110	

```
df['Timestamp'] = pd.to_datetime(df['Timestamp'])
df = df.sort_values(['City', 'Timestamp']).reset_index(drop=True)
```

```
df['jam'] = df['Timestamp'].dt.hour
df['hari'] = df['Timestamp'].dt.dayofweek
df['bulan'] = df['Timestamp'].dt.month
```

```
print(len(df))
```

```
254736
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 254736 entries, 0 to 254735
```

```
Data columns (total 16 columns):
#      Column      Non-Null Count  Dtype
---  -
0      Timestamp    254736 non-null  datetime64[ns]
1      City          254736 non-null  object
2      Latitude      254736 non-null  float64
3      Longitude     254736 non-null  float64
4      PM10_ug_m3     254710 non-null  float64
5      PM2_5_ug_m3    254736 non-null  float64
6      Carbon_Monoxide_ug_m3  254736 non-null  int64
7      Nitrogen_Dioxide_ug_m3  254736 non-null  float64
8      Ozone_ug_m3    254736 non-null  int64
9      Dust_ug_m3     254736 non-null  int64
10     UV_Index        254736 non-null  float64
11     European_AQI    254736 non-null  int64
12     Hazardous_Event 254736 non-null  int64
13     jam            254736 non-null  int32
14     hari            254736 non-null  int32
15     bulan           254736 non-null  int32
dtypes: datetime64[ns](1), float64(6), int32(3), int64(5), object(1)
memory usage: 28.2+ MB
```

2.Data CLeaning

```
print(df.isnull().sum()[df.isnull().sum()>0])
```

```
PM10_ug_m3    26
dtype: int64
```

```
print(len(df))
```

```
254736
```

```
df['PM10_ug_m3'] = df.groupby('City')['PM10_ug_m3'].transform(lambda x: x.fillna(x.median()))
```

```
print(len(df))
```

```
254736
```

```
print(df.isnull().sum())
```

```
Timestamp      0
City            0
Latitude        0
Longitude       0
PM10_ug_m3     0
PM2_5_ug_m3    0
Carbon_Monoxide_ug_m3  0
Nitrogen_Dioxide_ug_m3  0
Ozone_ug_m3    0
Dust_ug_m3     0
UV_Index       0
European_AQI   0
Hazardous_Event 0
jam            0
hari           0
bulan          0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

3.EDA

```
polusi_kota = df.groupby('City')['PM2_5_ug_m3'].mean().sort_values(ascending=False)

def tentukan_warna(nilai):
    if nilai > 75:
        return '#E24B4A' # Merah - Berbahaya
    elif nilai > 35:
        return '#EF9F27' # Kuning - Tidak Sehat
```

```

else:
    return '#378ADD' # Biru - Sedang

warna = [tentukan_warna(v) for v in polusi_kota.values]

# Buat visualisasi
fig, ax = plt.subplots(figsize=(10, 12))

bars = ax.barh(
    polusi_kota.index,
    polusi_kota.values,
    color=warna,
    edgecolor='none',
    height=0.7
)

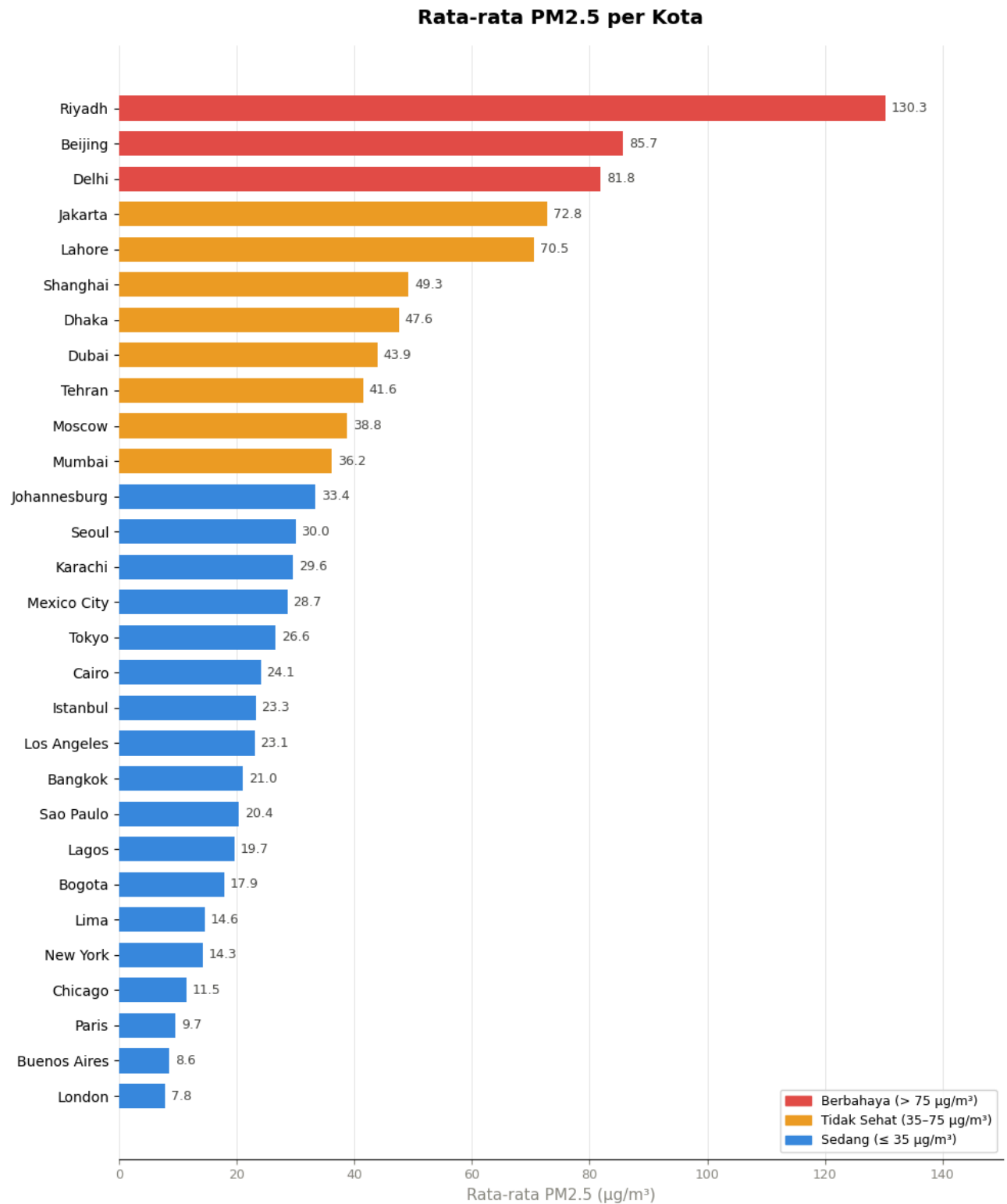
# Tambahkan label nilai di setiap bar
for bar, nilai in zip(bars, polusi_kota.values):
    ax.text(
        bar.get_width() + 1,
        bar.get_y() + bar.get_height() / 2,
        f'{nilai:.1f}',
        va='center',
        ha='left',
        fontsize=9,
        color='#444441'
    )

# Styling
ax.set_xlabel('Rata-rata PM2.5 (µg/m³)', fontsize=11, color='#888780')
ax.set_title('Rata-rata PM2.5 per Kota', fontsize=14, fontweight='bold', pad=15)
ax.invert_yaxis()
ax.set_xlim(0, polusi_kota.max() + 20)
ax.tick_params(axis='y', labelsize=10)
ax.tick_params(axis='x', labelsize=9, colors='#888780')
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.spines['left'].set_visible(False)
ax.xaxis.grid(True, color='#e0e0e0', linewidth=0.5)
ax.set_axisbelow(True)

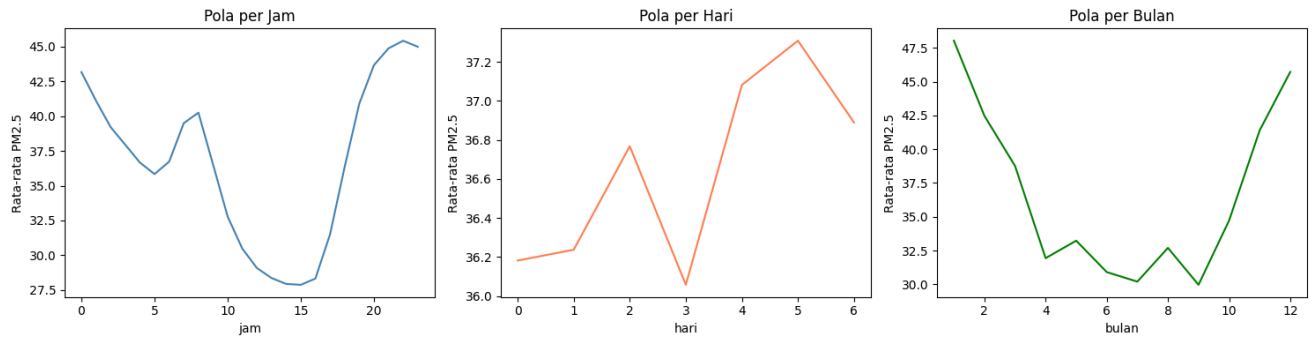
# Legend
legend_items = [
    mpatches.Patch(color='#E24B4A', label='Berbahaya (> 75 µg/m³)'),
    mpatches.Patch(color='#EF9F27', label='Tidak Sehat (35-75 µg/m³)'),
    mpatches.Patch(color='#378ADD', label='Sedang (≤ 35 µg/m³)')
]
ax.legend(
    handles=legend_items,
    loc='lower right',
    fontsize=9,
    framealpha=0.8
)

plt.tight_layout()
plt.savefig('visualisasi_polusi_kota.png', dpi=150, bbox_inches='tight')
plt.show()

```



```
# Visualisasi
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
df.groupby('jam')['PM2_5_ug_m3'].mean().plot(ax=axes[0], title='Pola per Jam', color='steelblue')
df.groupby('hari')['PM2_5_ug_m3'].mean().plot(ax=axes[1], title='Pola per Hari', color='coral')
df.groupby('bulan')['PM2_5_ug_m3'].mean().plot(ax=axes[2], title='Pola per Bulan', color='green')
for ax in axes:
    ax.set_ylabel('Rata-rata PM2.5')
plt.tight_layout()
```



4.Feature Engineering

```
df['is_weekend'] = (df['hari']>=5).astype(int)
df['is_workhour'] = (df['jam'].between(7,18)).astype(int)
print("Fitur Hari kerja dan jam Kerja telah Ditambahkan")
```

Fitur Hari kerja dan jam Kerja telah Ditambahkan

```
df['pm25_lag1'] = df.groupby('City')['PM2_5_ug_m3'].shift(1) # 1 jam lalu
df['pm25_lag3'] = df.groupby('City')['PM2_5_ug_m3'].shift(3) # 3 jam lalu
df['pm25_lag24'] = df.groupby('City')['PM2_5_ug_m3'].shift(24) # 24 jam lalu
```

```
print(len(df))
```

254736

Rolling mean 6 jam

Membuat fitur baru yang berisi rata-rata PM2.5 dari 6 jam terakhir untuk setiap kota. Tujuannya adalah menangkap tren jangka pendek — apakah polusi sedang naik atau turun dalam beberapa jam terakhir.

```
from pandas.core import window
df['pm25_rolling6'] = df.groupby('City')['PM2_5_ug_m3'].transform(lambda x:x.rolling(window=6,min_periods=1).mean())
```

```
df = pd.get_dummies(df,columns=['City'],prefix='city')
print("One hot encoding selesai")
```

One hot encoding selesai

```
print(len(df))
```

254736

```
df
```

	Timestamp	Latitude	Longitude	PM10_ug_m3	PM2_5_ug_m3	Carbon_Monoxide_ug_m3	Nitrogen_Dioxide_ug_m3	Ozone_ug_m3	Dust
0	2025-05-22 00:00:00	13.7563	100.5018	16.7	16.2	942	15.0	64	
1	2025-05-22 01:00:00	13.7563	100.5018	15.8	15.3	823	12.0	64	
2	2025-05-22 02:00:00	13.7563	100.5018	15.1	14.8	630	10.7	63	
3	2025-05-22 03:00:00	13.7563	100.5018	15.8	15.4	432	10.9	62	
4	2025-05-22 04:00:00	13.7563	100.5018	16.9	16.6	385	12.9	60	
...	...	...	...	...	...	...	...	...	...
254731	2026-05-22 19:00:00	35.6895	139.6917	13.1	11.1	221	18.4	67	
254732	2026-05-22 20:00:00	35.6895	139.6917	12.8	11.0	208	17.0	69	
254733	2026-05-22 21:00:00	35.6895	139.6917	12.2	10.6	196	16.0	70	
254734	2026-05-22 22:00:00	35.6895	139.6917	11.6	10.0	191	15.8	69	
254735	2026-05-22 23:00:00	35.6895	139.6917	11.4	10.0	187	16.1	68	

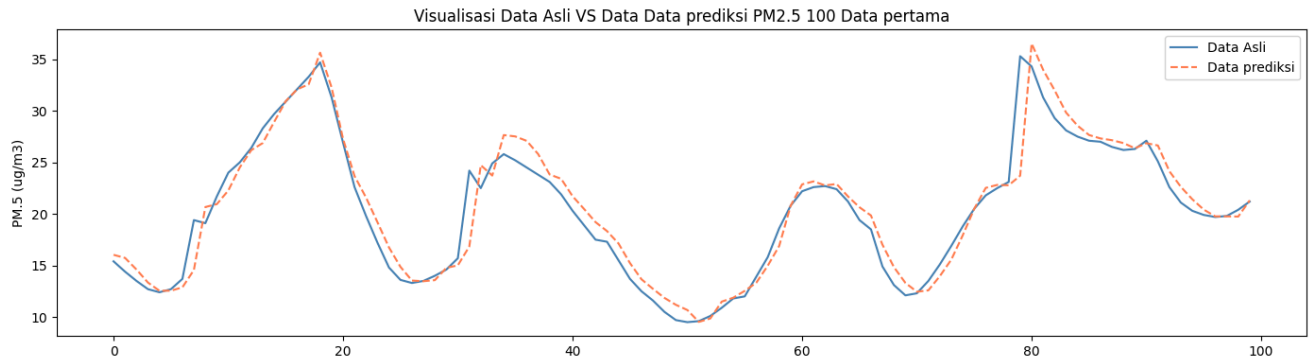
254736 rows × 50 columns

```
df.isnull().sum()
```


	0
Timestamp	0
Latitude	0
Longitude	0
PM10_ug_m3	0
PM2_5_ug_m3	0
print(len(df))	
254736	
Nitrogen_Dioxide_ug_m3	0
df = df.dropna()	
print(f'shape setelah feature engineering {df.shape}')	
shape setelah feature engineering (254040, 50)	
UV_Index	0
European_AQI	0
print(len(df))	
254040	
jam	0
batas_split = pd.Timestamp('2026-04-01')	
train = df[df['Timestamp'] < batas_split]	
test = df[df['Timestamp'] >= batas_split]	
train.shape	
(217848, 50)	
pm25_lag3	87
pm25_lag24	696
test.shape	
(36192, 50)	
city_Bangkok	0
column_drop = ['Timestamp', 'Latitude', 'Longitude', 'PM2_5_ug_m3', 'Hazardous_Event']	
fitur = [col for col in df.columns if col not in column_drop]	
city_Cairo	0
fitur	
['PM10_ug_m3',	
city_Delhi	0
'Carbon_Monoxide_ug_m3',	
city_Dhaka	0
'Nitrogen_Dioxide_ug_m3',	
city_Dhaka	0
'Ozone_ug_m3',	
city_Dubai	0
'Dust_ug_m3',	
city_Istanbul	0
'UV_Index',	
city_Istanbul	0
'European_AQI',	
city_Jakarta	0
'jam',	
city_Jakarta	0
'hari',	
city_Johannesburg	0
'is_weekend',	
city_Karachi	0
'pm25_lag1',	
city_Lagos	0
'pm25_lag3',	
city_Lagos	0
'pm25_lag24',	
city_Lahore	0
'pm25_lag24',	
city_Lima	0
'city_Bangkok',	
city_Lima	0
'city_Beijing',	
city_London	0
'city_Bogota',	
city_Los_Angeles	0
'city_Los_Angeles',	
city_Mexico_City	0
'city_Chicago',	
city_Mexico_City	0
'city_Dhaka',	
city_Moscow	0
'city_Dubai',	
city_Mumbai	0
'city_Istanbul',	
city_Jakarta	0
'city_Johannesburg',	
city_New_York	0
'city_Karachi',	
city_Paris	0
'city_Lahore',	
city_Riyadh	0
'city_Riyadh',	
city_Sao_Paulo	0
'city_London'	
city_Sao_Paulo	0

Jumlah fitur yang digunakan 45

```
plt.figure(figsize=(14, 4))
plt.plot(y_test.values[:100],label="Data Asli",color='steelblue')
plt.plot(y_pred[:100],label='Data prediksi',color='coral',linestyle='--')
plt.title("Visualisasi Data Asli VS Data Data prediksi PM2.5 100 Data pertama")
plt.ylabel('PM.5 (ug/m3)')
plt.legend()
plt.tight_layout()
```



```
df_perbandingan = pd.DataFrame({
    'Data Asli' : y_test.values,
    'Data Prediksi' : y_pred,
    'Selisih ' : (y_test.values - y_pred).round(2),
    'Selisih %' : (abs(y_test.values - y_pred) / y_test.values * 100).round(2)
})

df_perbandingan.head(10)
```

```
/tmp/ipykernel_903/4240722884.py:5: RuntimeWarning: divide by zero encountered in divide
'Selisih %' : (abs(y_test.values - y_pred) / y_test.values * 100).round(2)
```

	Data Asli	Data Prediksi	Selisih	Selisih %
0	15.4	16.032086	-0.63	4.10
1	14.4	15.731957	-1.33	9.25
2	13.5	14.604799	-1.10	8.18
3	12.7	13.338546	-0.64	5.03
4	12.4	12.542727	-0.14	1.15
5	12.7	12.552602	0.15	1.16
6	13.7	12.877213	0.82	6.01
7	19.4	14.558040	4.84	24.96
8	19.1	20.668559	-1.57	8.21
9	21.7	20.948928	0.75	3.46

```
print(f'{df_perbandingan["Selisih %"].mean():.2f}%')
```

```
inf%
```

```
range_data = y_test.max() - y_test.min()
print(f'range datanya adalah {range_data}')
```

```
range datanya adalah 588.6
```

7.Importance Fitur

Klik dua kali (atau tekan Enter) untuk mengedit

```
#mencari fitur mana yang paling berpengaruh
importance_fitur = pd.Series(model.feature_importances_, index=fitur).sort_values(ascending=False)
print("Fitur Fitur Yang paling berpengaruh")
print(importance_fitur.head(10).round(4))
```

```
Fitur Fitur Yang paling berpengaruh
pm25_lag1          0.9537
PM10_ug_m3         0.0357
Dust_ug_m3         0.0053
Carbon_Monoxide_ug_m3 0.0015
pm25_lag3         0.0010
pm25_rolling6     0.0006
city_Moscow        0.0006
Nitrogen_Dioxide_ug_m3 0.0005
European_AQI       0.0002
city_Istanbul      0.0002
dtype: float64
```

```
#membuat fitur paling berpengaruh dalam visualisasi
plt.figure(figsize=(10,4))
importance_fitur.head(10).plot(kind='bar',color='steelblue')
plt.title("Top 10 Fitur yang paling berpengaruh")
plt.ylabel("Importance Score")
plt.xticks(rotation=45,ha='right')
plt.tight_layout()
plt.show()
```

